

# Statistics Homework 1:

## Introduction

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### Note regarding z-tables

Some items on this homework will require you to look up probabilities on a z-table. A copy of a z-table is provided on Canvas (in the document titled “Tables”). Whenever a z-table is needed to answer a question on a learning assessment, the required portion of the z-table will be provided and displayed with the question.

### Remember: Do not use a percent sign

When entering answers on Canvas, remember that Canvas does NOT recognize or understand percentage signs. Therefore, if a numerical answer involves a percentage, type the answer using a decimal and not a percentage sign. For example, if the answer is 28 percent, DO NOT type “28%,” but rather, type “.28” when entering your answer.

## Part A: Definitions

Match the term with the definition.

1. A numerical value based on data from a **sample** that indicates something about a **population**.
2. The term used when the expected value is not equal to the parameter.
3. A numerical value based on data from a **sample** that indicates something about the **sample**.
4. The average of all the values for a population of values from which a single value can be randomly sampled.
5. The general term for a numerical value based on data from a sample (that may indicate something about the sample or something about the population).
6. The set of cases from which a researcher collects data.
7. A numerical value based on data from **all the cases in a population** that describes something about the population.
8. All of the cases in which a researcher is interested.
9. The same amount of something always produces the same assessed numerical value regardless of when or how it is assessed.
10. The extent to which a scale measures what it is supposed to measure.
11. Study where scores for all the variables are determined by assessment and there are NO manipulated variables.
12. Study that has at least one variable with scores determined by an assessment, and at least one manipulated variable produced by the researcher forcing cases into groups.

- |                           |                          |
|---------------------------|--------------------------|
| A. Bias                   | G. Descriptive statistic |
| B. Reliability            | H. Statistic             |
| C. Population             | I. Validity              |
| D. Expected value         | J. Experimental research |
| E. Sample                 | K. Inferential statistic |
| F. Correlational research | L. Parameter             |

## Part B: Saying “you” and “I” during conflict

In some approaches to couple therapy, partners are given guidelines for how to talk to each other when they have relationship conflicts. Sometimes they are instructed to reduce statements that criticize and blame each other, and instead, to increase statements that talk about one’s own personal perceptions, feelings, and desires. These different types of statements can be assessed by counting different types of pronouns that

people use. Statements that blame a partner typically contain the word “you” whereas statements about the self often contain the word “I.”

Simmons, Gordon, and Chambless (2005) videotaped a sample of 59 married couples as they discussed an area of relationship conflict. The recordings were later transcribed, and a computer program was used to count the frequency of different words. This included: (1) counting the number of times a person used a pronoun, (2) counting the number of times a person used the pronoun “you,” and (3) counting the number of times a person used the pronoun “I.” Two scores were then calculated for each participant. The first score was the percent of a participant’s pronouns that used the word “you.” The second score was the percent of a participant’s pronouns that used the word “I.”

Scores regarding the percent of “you” pronouns from 9 participants are listed below. Each score indicates the percent of pronouns that use the word “you” for a given person. Compute the mean, the sum of squares, the variance, and the standard deviation. Although the sample depicted here includes only 9 cases, if you do your calculations correctly, your results will be identical to the results reported by Simmons, Gordon, and Chambless (2005).

Percent of pronouns using the word “you”

|          | score |
|----------|-------|
| Person 1 | .19   |
| Person 2 | .38   |
| Person 3 | .41   |
| Person 4 | .14   |
| Person 5 | .16   |
| Person 6 | .40   |
| Person 7 | .22   |
| Person 8 | .36   |
| Person 9 | .44   |

Compute the following

13. Mean =

14. Sum of squares =

15. Variance =

16. Standard deviation =

You just calculated some descriptive statistics for the percent of pronouns using the word “you.” The SPSS output below is **based on a larger sample**, and it provides descriptive statistics for the percent of pronouns using the word “I.” From this output, determine the variance and the sum of squares. Also answer the questions regarding z-scores and probability.

### Descriptive Statistics

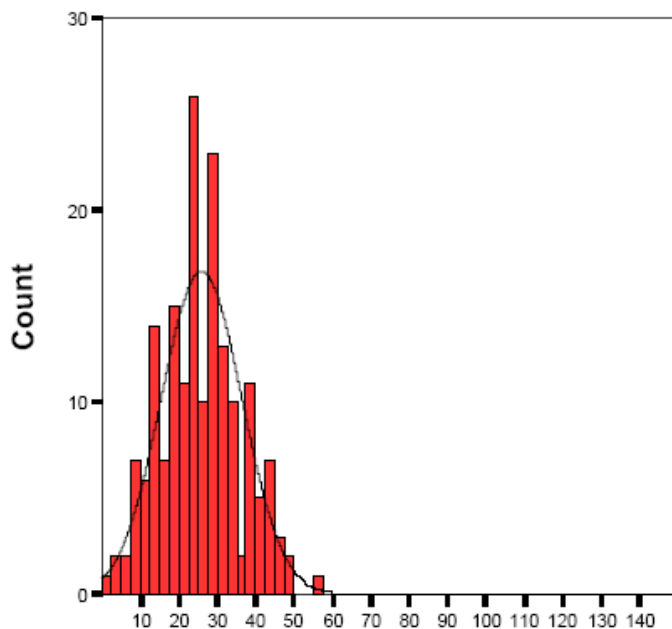
|                    | N   | Minimum | Maximum | Mean | Std. Deviation |
|--------------------|-----|---------|---------|------|----------------|
| Percent "I"        | 118 | .08     | .72     | .43  | .14            |
| Valid N (listwise) | 118 |         |         |      |                |

17. For the percent of pronouns using “I,” what is the variance?
18. For the percent of pronouns using “I,” what is the sum of squares?
19. If Rachel receives a score of .72 (that is, she uses the word “I” in .72 of her pronouns), what is her z-score?
20. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score higher than Rachel?
21. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score more extreme than Rachel? (Note: “more extreme” means further from the mean in either direction.)
22. If Peter receives a score of .38, what is his z-score?

23. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score lower than Peter?
24. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score more extreme than Peter? (Note: “more extreme” means further from the mean in either direction.)

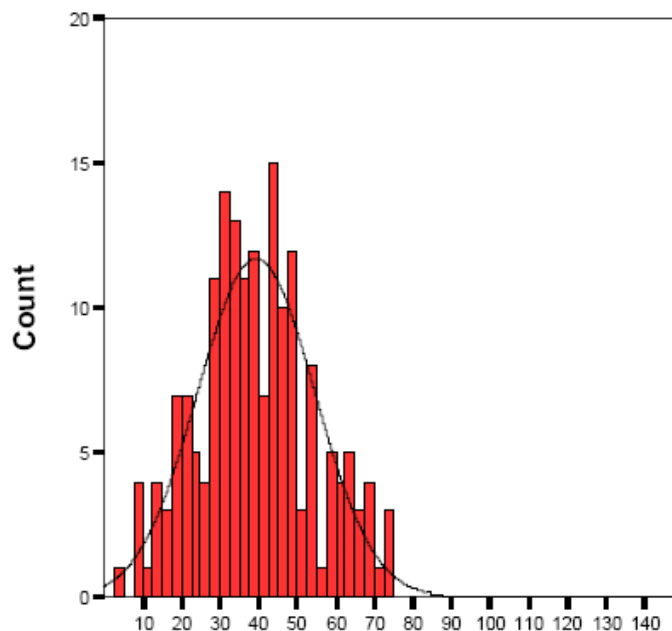
### Part C: Visually estimating standard deviations

By now you should have a strong understanding of what a standard deviation is. You should be able to provide a good estimate of the standard deviation just by looking at a histogram display of data from a study. Look at the following histograms, and do NOT make any calculations. Simply look at the histogram and provide a rough approximation of the standard deviation. You should then be able to judge which of the options is correct.



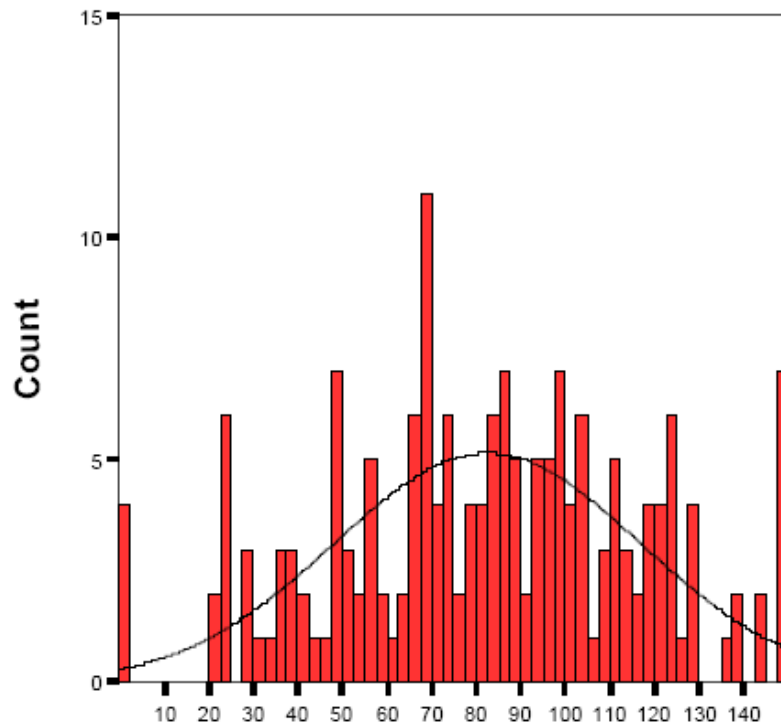
25. In this histogram, the standard deviation appears to be about:

- A. 4
- B. 11
- C. 26
- D. 36
- E. 60



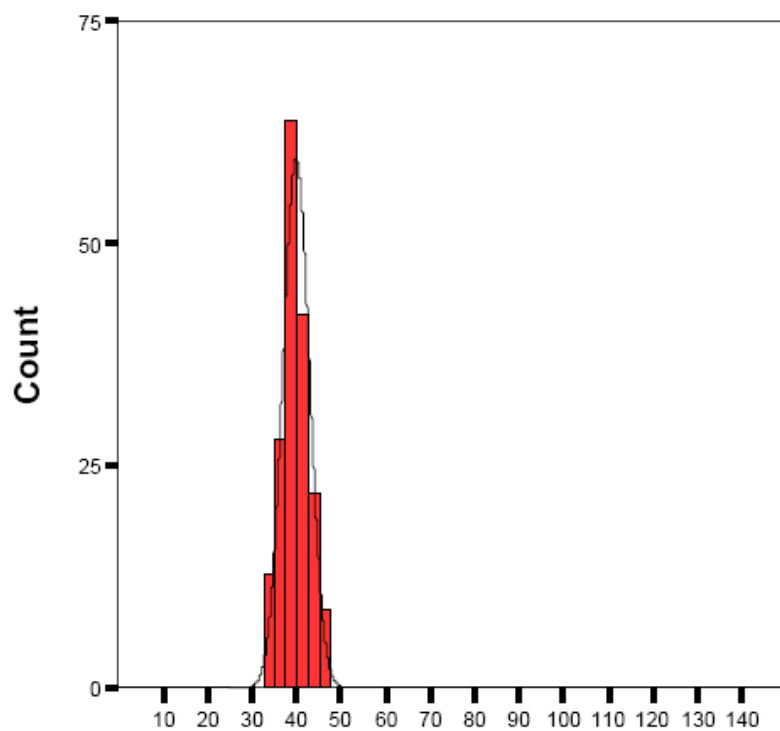
26. In this histogram, the standard deviation appears to be about:

- A. 5
- B. 15
- C. 26
- D. 38
- E. 90



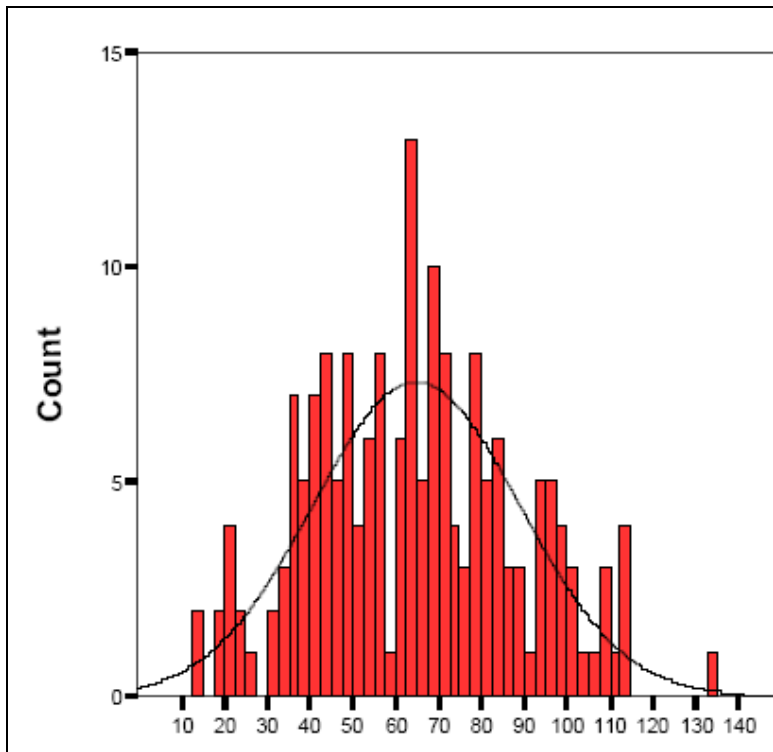
27. In this histogram, the standard deviation appears to be about:

- A. 10
- B. 14
- C. 34
- D. 60
- E. 140



28. In this histogram, the standard deviation appears to be about:

- A. 3
- B. 10
- C. 14
- D. 26
- E. 40



29. In this histogram, the standard deviation appears to be about:

- A. 9
- B. 14
- C. 24
- D. 70
- E. 130

#### Part D: Frequency of conflict and support in dating couples

Campbell, Simpson, Boldry, & Kashy (2005) conducted a study investigating frequency of conflict and support in dating relationships. They used a sample of 103 dating couples (103 men and 103 women) attending Texas A&M University. Each member of each couple kept a daily diary for 14 days. On each day, participants recorded whether they had a conflict interaction with their partner, and whether they experienced a supportive event with their partner.

Conflict scores from 12 participants are listed below. Each score indicates the number of days that a person reported conflict out of 14 days total. Compute the mean, the sum of squares, the variance, and the standard deviation. Although the sample depicted here includes only 12 cases, if you do your calculations correctly, your results will be identical to the results reported by Campbell, Simpson, Boldry, & Kashy (2005).

Number of days experiencing conflict (out of 14).

|           | score |
|-----------|-------|
| Person 1  | 7     |
| Person 2  | 6     |
| Person 3  | 0     |
| Person 4  | 11    |
| Person 5  | 5     |
| Person 6  | 13    |
| Person 7  | 3     |
| Person 8  | 8     |
| Person 9  | 4     |
| Person 10 | 10    |
| Person 11 | 11    |
| Person 12 | 6     |

30. Mean =

31. Sum of squares =

32. Variance =

33. Standard deviation =

You just calculated some descriptive statistics for the number of days participants experienced conflict. The SPSS output below is based on a larger sample, and it provides descriptive statistics for the number of days

participants reported experiencing social support. As with the conflict scores, the social support scores indicate the number of days that a person reported experiencing social support out of 14 days total. From this output, determine the variance and the sum of squares. Also answer the questions regarding z-scores and probability.

### Descriptive Statistics

|                    | N   | Minimum | Maximum | Mean  | Std. Deviation |
|--------------------|-----|---------|---------|-------|----------------|
| days support       | 206 | .0      | 14.0    | 10.39 | 3.00           |
| Valid N (listwise) | 206 |         |         |       |                |

34. For the number of days experiencing support, what is the variance?
35. For the number of days experiencing support, what is the sum of squares?
36. If Loren experiences social support on 5 days, what is her z-score?
37. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score lower than Loren?
38. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score more extreme than Loren? (Note: "more extreme" means further from the mean in either direction.)
39. If Jeff experiences social support on 12 days, what is his z-score?
40. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score higher than Jeff?
41. Assuming that scores are normally distributed, what is the probability that a person drawn from this sample will get a score more extreme than Jeff? (Note: "more extreme" means further from the mean in either direction.)

### Part E: Memorized area under the curve

You should have memorized the following areas under the normal distribution. You may report values that are rounded just as they were in class lecture.

42. In a normal distribution, what percent of cases fall between a score that is at the mean and a score that is one standard deviation above the mean?
43. In a normal distribution, what percent of cases fall between a score that is one standard deviation above the mean and a score that is two standard deviations above the mean?
44. In a normal distribution, what percent of cases fall above the second standard deviation?

### Part G: Estimating $\hat{x}_i$ and $e_i$

The dataset to the right includes only four cases. Calculate  $\hat{x}_i$  and  $e_i$  for each case.

45. What is  $\hat{x}_i$  for Noah?
46. What is  $\hat{x}_i$  for Oswald?
47. What is  $\hat{x}_i$  for Peter?
48. What is  $\hat{x}_i$  for Quinn?
49. What is  $e_i$  for Noah?
50. What is  $e_i$  for Oswald?
51. What is  $e_i$  for Peter?
52. What is  $e_i$  for Quinn?

|        | X  | $\hat{x}_i$ | $e_i$ |
|--------|----|-------------|-------|
| Noah   | 11 |             |       |
| Oswald | 9  |             |       |
| Peter  | 13 |             |       |
| Quinn  | 7  |             |       |

## Part H: Questions regarding a linear equation and a frequency distribution

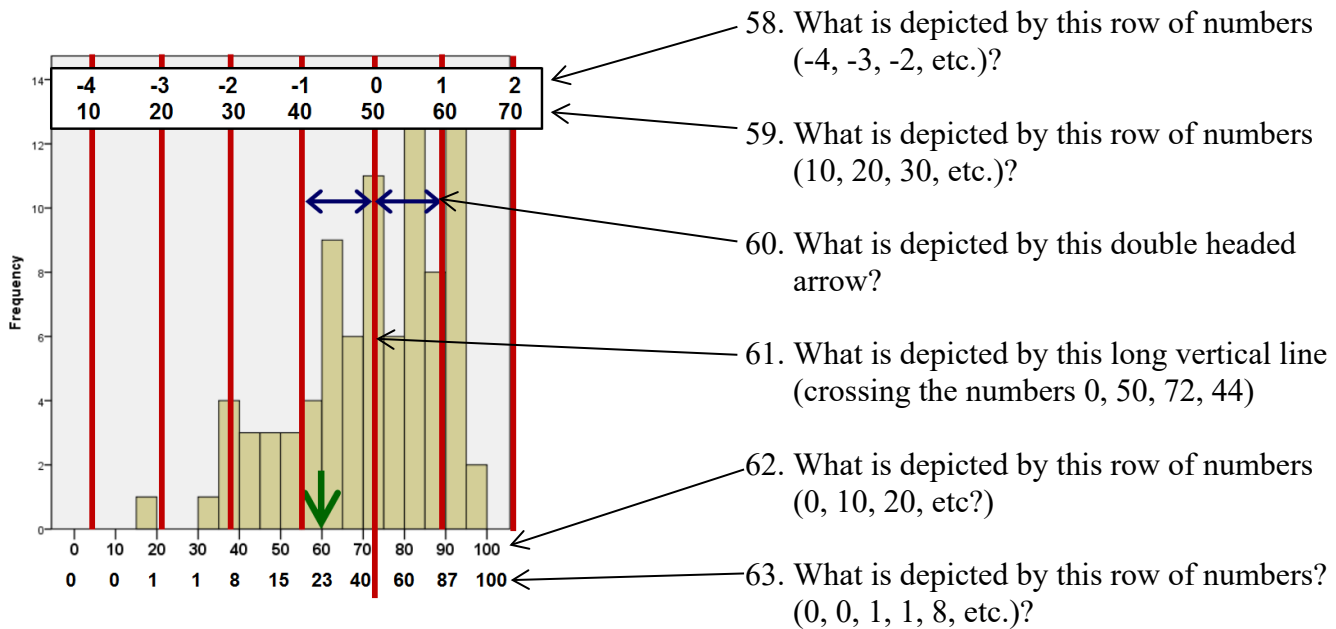
Use the list of options below to answer each of the questions in this section. Some options may be used more than once.

|                                             |                                  |             |                              |
|---------------------------------------------|----------------------------------|-------------|------------------------------|
| A. how much case $i$ deviates from the mean | C. T-scores                      | F. Zero     | H. standard deviation        |
| B. $\bar{x}$                                | D. The score on $x$ for case $i$ | G. z-scores | I. Unstandardized raw scores |
|                                             | E. Percentiles                   |             |                              |

The questions below pertain to the equation to the right:

53. What is  $x_i$ ?
54. What value is  $\hat{x}_i$  equal to?
55. What is  $e_i$ ?
56. What is the mean of  $e_i$ ?
57. What commonly calculated statistic is essentially the mean of  $|e_i|$ ?

$$x_i = \bar{x} + e_i$$



## Part I: Written summaries

The “Flourishing Scale” is a questionnaire that measures the extent to which people feel like they are flourishing in life. Respondents use a 7-point scale to rate 8 different questions about life satisfaction and life quality. A total score is created by summing all 8 questions. Descriptive statistics for this scale are printed below.

### Descriptive Statistics

|                    | N   | Minimum | Maximum | Mean    | Std. Deviation |
|--------------------|-----|---------|---------|---------|----------------|
| FS_8               | 482 | 8.00    | 56.00   | 43.2220 | 9.41084        |
| Valid N (listwise) | 482 |         |         |         |                |

64. **Write a summary of the mean and the standard deviation of the Flourishing scale.** Your written summary should follow the template provided in your notes and described in the lecture add-on video. After you have provided a written summary, use the scoring system in the box below to grade your own answer. On Canvas, you will be asked to report if you wrote this summary and graded your own answer. The scoring system in the box below is the same as the scoring system that will be used to grade written summaries on learning assessments.

**Scoring for a written summary of a mean and standard deviation for a variable**

This question is worth 2 points. Make deductions as specified below with a maximum deduction of 2 points.

Deduct one point if any of the following components are not included:

- The summary needs to name the scale being summarized. In this case, the scale should be called the “Flourishing Scale.” The name needs to be stated in plain language, and the summary should NOT contain any cryptic code names such as “FS\_8.”
- The summary needs to state that scores on the scale can range from 8 to 56. (The summary should NOT describe this range merely as a “7-point scale,” because this is ambiguous and not actually correct.)
- The summary needs to state that the mean is 43.22.
- The summary needs to make it clear that you understand the meaning of a standard deviation. Specifically, the summary needs to include a statement indicating that, on average, people deviate 9.41 units from the mean.
- The summary needs to use the correct value for the standard deviation (which is 9.41).

Deduct one-half point for problems in any of the following areas:

- To make written summaries easy to read values should be rounded to two decimal places.

Below is an example of an answer that would receive full credit:

On a Flourishing Scale that ranged from 8 to 56, the average score was 43.22. On average, people deviated from this score by 9.41 units.

65. Descriptive statistics for the Flourishing Scale are printed above. Sam received a score of 55 on this scale. Calculate Sam’s z-score and **write a summary of Sam’s z-score.** Your written summary should follow the template provided in your notes and described in the lecture add-on video. After you have provided a written summary, use the scoring system in the box below to grade your own answer. On Canvas, you will be asked to report if you wrote this summary and graded your own answer. The scoring system in the box below is the same as the scoring system that will be used to grade written summaries on learning assessments.

**Scoring for a written summary of a person’s z-score**

This question is worth 2 points. Make deductions as specified below with a maximum deduction of 2 points.

Deduct one point if any of the following components are not included:

- The summary needs to make it clear that you understand the meaning of a z-score. In this case, the summary needs to include a statement indicating the person is 1.25 standard deviations above the mean. This interpretation must include the words “standard deviations,” and not something ambiguous like “units.”

- The summary needs to use the correct value for the z-score. In this case, you should have determined that the value is 1.25.
- The summary needs to indicate that, if scores are normally distributed, Sam's score falls between the 84<sup>th</sup> and the 98<sup>th</sup> percentile.

Deduct one-half point for problems in any of the following areas:

- To make written summaries easy to read values should be rounded to two decimal places.
- Only deduct one half point if the statement about the percentile range omits the part about "if scores are normally distributed."

Below is an example of an answer that would receive full credit:

Sam's flourishing score was 1.25 standard deviations above the mean. If flourishing is normally distributed, Sam's score is between the 84<sup>th</sup> and the 98<sup>th</sup> percentile.

## References

- Campbell, L., Simpson, J. A., Boldry, J., & Kashy, D. A. (2005). Perceptions of Conflict and Support in Romantic Relationships: The Role of Attachment Anxiety. *Journal of Personality and Social Psychology*, 88, 510-531.
- Simmons, R. A., Gordon, P. C., & Chambless, D. L. (2005). Pronouns in marital interaction: What do 'you' and 'I' say about marital health? *Psychological Science*, 16, 932-936.